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```
clear;
clc;
close all;
```

Load Talley's spark-only test file

```
data = load('SPARK_ONLY_001.mat');

iq = data.eventIQ;
fs = data.fs;
fc = data.fc;
```

Basic information

```
fprintf('Samples: %d\n', length(iq));
fprintf('Sample rate: %.0f Hz\n', fs);
fprintf('Center frequency: %.1f MHz\n', fc/1e6);
```

```
Samples: 2400000
Sample rate: 2400000 Hz
Center frequency: 145.9 MHz
```

Time axis

```
t = (0:length(iq)-1) / fs;
```

I/Q magnitude

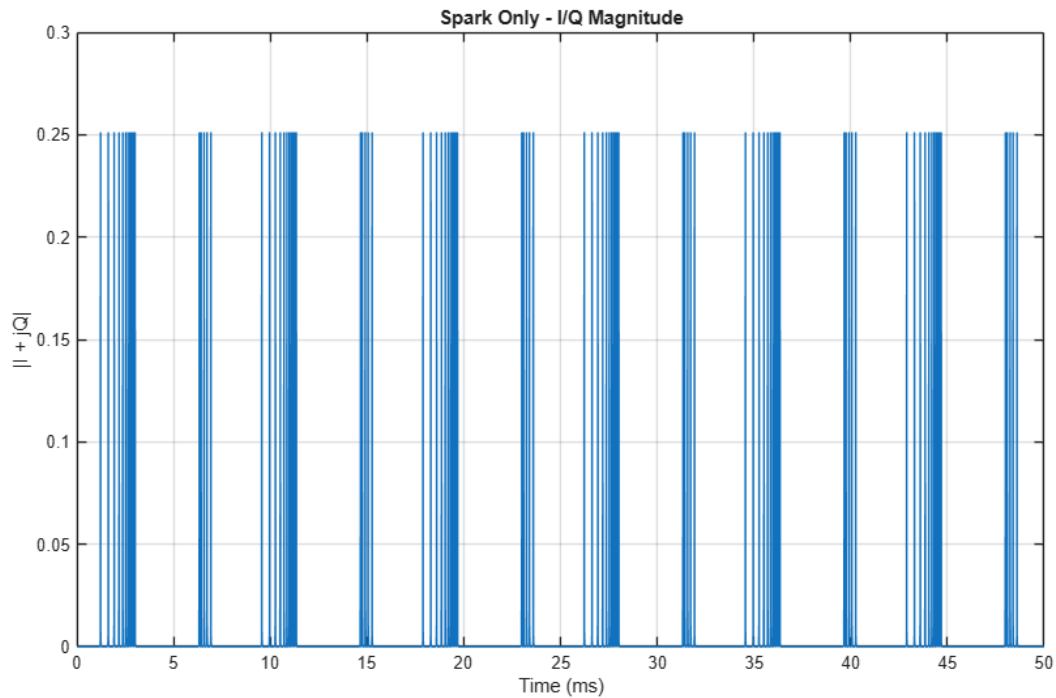
```
mag = abs(iq);
```

Plot first 50 ms

```
showTime = 0.05;
Nshow = round(showTime * fs);

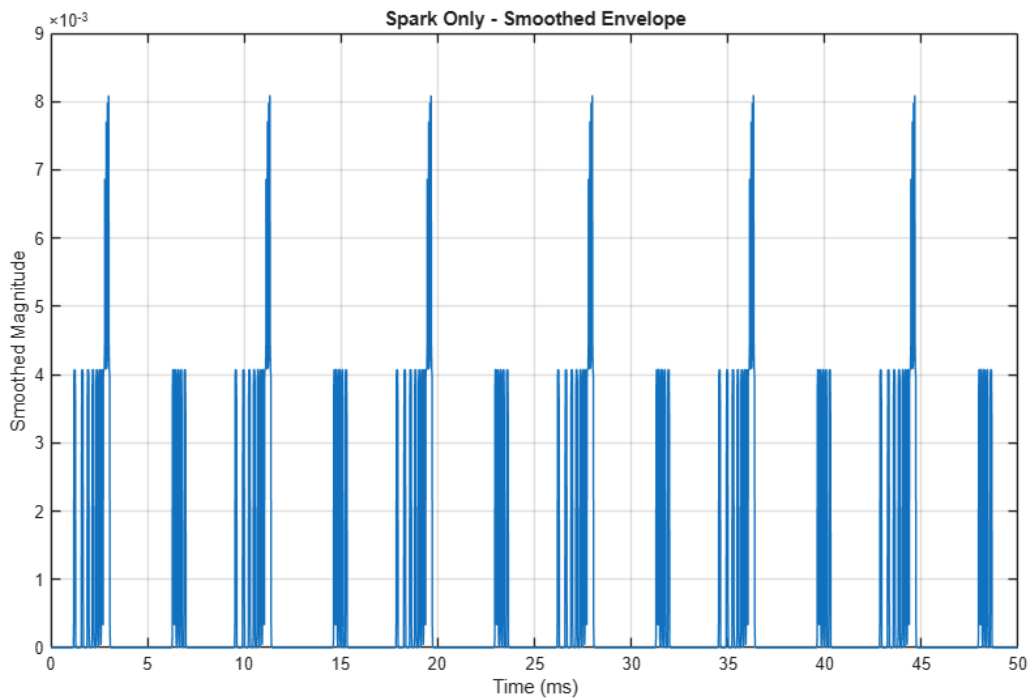
figure;
```

```
plot(t(1:Nshow)*1000, mag(1:Nshow));  
xlabel('Time (ms)');  
ylabel('|I + jQ|');  
title('Spark Only - I/Q Magnitude');  
grid on;
```



Simple smoothing

```
windowLength = 200;  
smoothMag = movmean(mag, windowLength);  
  
figure;  
plot(t(1:Nshow)*1000, smoothMag(1:Nshow));  
xlabel('Time (ms)');  
ylabel('Smoothed Magnitude');  
title('Spark Only - Smoothed Envelope');  
grid on;
```



Basic measurements

```
meanLevel = mean(mag);  
rmsLevel = rms(mag);  
peakLevel = max(mag);  
  
fprintf('\nMean magnitude: %.6f\n', meanLevel);  
fprintf('RMS magnitude: %.6f\n', rmsLevel);  
fprintf('Peak magnitude: %.6f\n', peakLevel);  
fprintf('Peak / RMS ratio: %.2f\n', peakLevel/rmsLevel);
```

```
Mean magnitude: 0.000693  
RMS magnitude: 0.008422  
Peak magnitude: 0.251189  
Peak / RMS ratio: 29.83
```

Published with MATLAB® R2026a